Faculty of Computer Science & Information Technology

GROUP PROJECT (30%)
WIA2006 SYSTEMS ANALYSIS, MODELLING & DESIGN

1. Overview & Motivation

In software development, high-quality code depends entirely on thorough analysis and system design. This project provides a hands-on experience in the System Development Life Cycle (SDLC) by challenging you to solve a real-world operational or business friction point within the Faculty of Computer Science and Information Technology (FCSIT), Universiti Malaya.

Working in teams, you will experience the important phases involved in SDLC from initial domain investigation and requirements elicitation to system architecture modeling, UI/UX design, and functional prototyping. This initiative bridges abstract theoretical methodologies with practical software development practices.

2. Learning Objectives

By completing this project, students will be able to:

- Identify, analyze, and formalize functional and non-functional requirements by interviewing or surveying real campus stakeholders (FCSIT students, academic staff, and administrative personnel).
- Apply standard techniques (e.g., Object-Oriented Analysis & Design using UML, Data Flow Diagrams, or Use Case Analysis) to model complex system behaviors, data flows, and structural relationships.
- Identify inefficiencies in existing manual, legacy, or informal workflows and propose an automated, scalable software solution.
- Translate architectural models into a functional, user-centric mobile application or web application prototype.
- Produce an industry-standard System Requirements & Design Specification (SRDS) report.

3. Project Scope & Deliverables

Scope Constraints

- **Team Structure:** Form a group of exactly **five (5) students**.
- **Target Domain:** The proposed system **must specifically target FCSIT staff, faculty members, or students** to improve existing business, academic, or operational processes.
- **Platform:** Mobile Application (Android/iOS) or Web Application.

Core Deliverables

1. **System Requirements & Design Report:** A comprehensive document detailing stakeholder analyses, problem statements, current system reviews, architectural diagrams, data schema/models, and UI/UX wireframes.

2. **Interactive Prototype:** A working, functional prototype (web or mobile) demonstrating core features and user workflows.

4. Project Phases & Methodology

1. **Phase 1: Domain Investigation & Analysis (As-Is System)**
    - Conduct empirical research on current mechanisms or workflows used within FCSIT (e.g., lab room bookings, thesis submissions, consultation scheduling, course resource management).
    - Perform root-cause analysis to pinpoint operational bottlenecks, data redundancies, or user pain points.

2. **Phase 2: Proposed System Design (To-Be System)**
    - Architect an enhanced system or a new solution.
    - Construct structural models (e.g., Class Diagrams, ER Diagrams) and behavioral models (e.g., Use Case Diagrams, Sequence Diagrams).
    - Define software architecture, interface design, and security/access control requirements.

3. **Phase 3: Prototype Implementation & Validation**
    - Develop a functional prototype showcasing core operational flows.
    - Evaluate the prototype against user requirements to validate usability and alignment with initial design specifications.

5. Suggested Focus Areas & System Domains

To kickstart your brainstorming, consider addressable problem domains within FCSIT:

- **Academic Operations:** Automated consultation hour scheduling, FYP/Master's thesis repository & tracking, peer code-review platforms.
- **Faculty Resources:** Smart lab equipment checkout system, real-time study space/lab availability tracker, facility maintenance reporting.
- **Student Life & Engagement:** Intra-faculty event discovery and attendance verification, peer tutoring/mentorship pairing, student society asset management.

Structure of Report:

1. Introduction
2. Project Roles.
3. Budget
4. Project Timeline
5. Planning
6. Analysis
7. Design
8. Implementation
9. System Evaluation and Testing
10. References
11. Appendix
    a. Questionnaires/interview Scripts
    b. Fact finding Result/Existing System/Form/Picture

Front page: Title of project and a list of group members

Introduction – Overview of the system.

Planning

- Fact Finding Conducted (Interview/Observation)
- Existing Systems Description (Existing System FDD (Functional Decomposition Diagram)
- List the problem/limitation
- List the suggestion
- Proposed System FDD (with the suggestions above)

Analysis

Data and Process Modeling

- DFD Diagrams (Context Diagram, Diagram 0 DFD)
- Data Dictionary (Sample only)
- Includes the description

Design and Implementation

- **Design**

- Database design: ERD Diagram**,** Use Case diagram. There is no need to implement the database.

- **Implementation -** Screen shots of User Interface Design based on diagrams.

- **Evaluation/Testing**

Project Roles

Position	Name	Task
Project Manager		
Programmer		
System Analyst		
Network Engineer		

Budget

Item	Amount	Total (RM)
Server	RM600 per year	1200
Salary/ Manpower	RM3000 x 5 members	15000
		16200

Submission Requirement:

Project Deliverables

Due Date	Deliverables
5- 14 Jan 2027	Presentation/Final Report Draft
5Jan 2027	Final Report Submission

Assessment Criteria:

Assessment Criteria	Marks
Report	20%
Introduction	1
Project Roles/Budget/Timeline	2
Planning (Fact Finding/Limitation/Recommendation/FDD Existing)	3
Planning (FDD Proposed)	3
Analysis (Context Diagram/DFD)	3
System Design (ERD/Use Case Diagram)	2
Implementation	2
System Evaluation & Testing	2
Reference/Appendix	2
Prototype	10%
User Interface	7
Group Work/Presentation	3
Total Percentage	30%